Ilminur Abulikemu

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EDUCATION

Middle Tennessee State University Aug. 2025 – Present
M.S. in Data Science, GPA: 3.95/4.0 *Murfreesboro, TN*
- Coursework: Data Mining, Predictive Modeling, Applied Statistical Methods, Model Optimization & Deployment
- Award: MTSU International Merit Scholarship (2025–2026)

Shanghai University of Finance and Economics Sep. 2021 – Jun. 2025
B.A. in Data Journalism, GPA: 3.53/4.0 *Shanghai, China*
- Coursework: Computer Programming, Calculus, Data Mining, Statistical Software, Business Data Analysis
- Award: SUFE Inspirational Scholarship (three consecutive years, 2021–2024)

PUBLICATIONS

Zhang, H., **Abulikemu, Y.**, & Zhou, W. (2026). Measurement and Classification of Piezoelectric Actuator Motion Using Heterodyne Interferometry and Large Language Models. *ASME MSEC 2026*.

RESEARCH EXPERIENCE

Research Assistant, Smart Sensing and Robotics Lab Aug. 2025 – Present
Middle Tennessee State University
- Developing a physics-informed digital twin of an open-loop piezoelectric actuator that couples calibrated hysteresis modeling with a Mamba-based network trained to learn and correct the physics model's residual error
- Designed a custom LLM framework and a Transformer-based classifier that classify actuator motion from heterodyne-interferometry measurements with nanometer-level precision
- Built the end-to-end data pipeline (preprocessing, database management, cross-validation, hyperparameter tuning) and a closed-loop Python PID controller that turns diagnostic outputs into control improvements

Research Intern, Guo Lab Aug. 2026 – Present
Vanderbilt University Medical Center
- Curated and annotated a reference set of 20 human TF motif profiles from JASPAR (CORE, *Homo sapiens*) with matrix IDs, TF families, and MEME-format matrices as input to a TF-informed TWAS (sTF-TWAS) study
- Processing epithelial scATAC-seq peak data (GRCh38; BEDTools, samtools) to identify candidate TF-binding sites

Research Assistant, Neuromarketing & AI Lab Jan. 2026 – Present
Middle Tennessee State University
- Built a data-processing workflow that merges raw per-participant reaction-time files (Testable) across experimental conditions into analysis-ready datasets covering 700+ participants
- Established participant-tracking and quality-control procedures to ensure traceable, reproducible data

OTHER EXPERIENCE

Data Science Intern, The Journey Home (Non-Profit) Aug. 2026 – Present
- Designing a database-backed intake platform to replace paper-based client intake at a homeless-services nonprofit
- Developed a normalized relational data model, a privacy-based data-sensitivity framework, and a phased rollout plan

Graduate Assistant, Jones College of Business IT Services, MTSU May 2026 – Present
- Build and maintain 10+ Power BI dashboards for cross-departmental reporting; develop Python/JS automation tools

PROJECTS

AscendFi | *Python, FastAPI, Claude API, Vue/Nuxt, Supabase* — VandyHacks Mar. 2026
- Built a full-stack AI finance platform with a multi-agent system for risk prediction and debt optimization

Proto Space | *Unity, Meta Quest 3* — HackMT Jan. 2026
- Developed an augmented-reality CAD visualizer with hand-gesture interaction for inspecting imported 3D models

TECHNICAL SKILLS

Programming	Python (PyTorch, scikit-learn, pandas), R, SQL, JavaScript, Git
ML & Analytics	Transformers, Mamba, LLMs, NLP, Physics-Informed Modeling, Statistical Analysis, Power BI, Tableau
Languages	English (Fluent), Mandarin (Native), Uyghur (Native), Korean (Intermediate)